

Worksheet 12

October 21, 2026

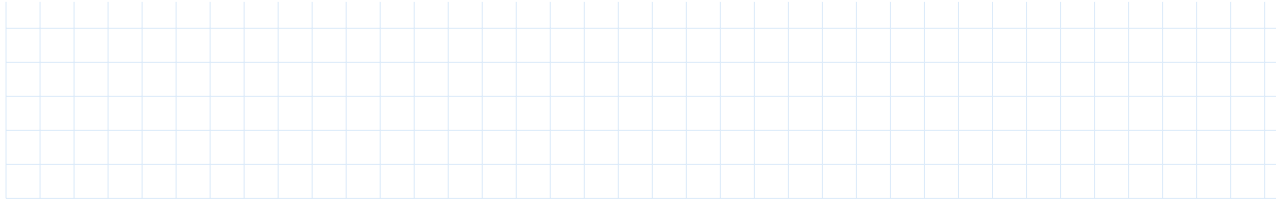
Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

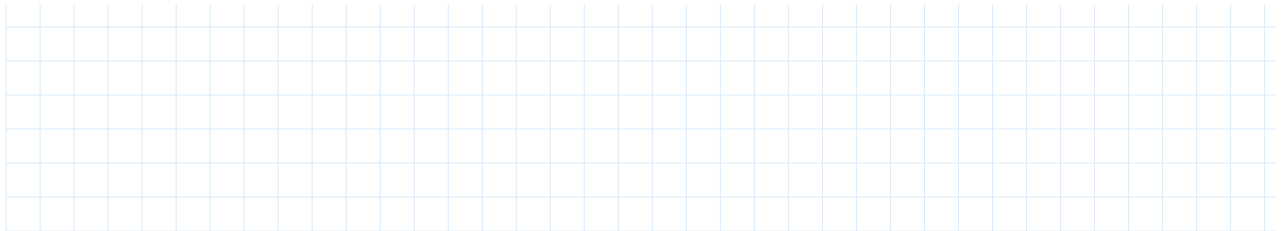
1. Radians, arc, and sector

A sector has radius 6 cm and central angle 150° .

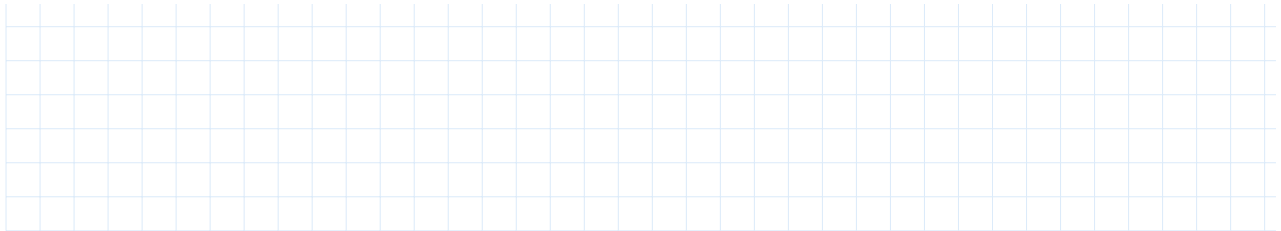
(a) Convert the angle to radians.



(b) Find the exact arc length.



(c) Find the exact area of the sector.



Worksheet 12

October 21, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

2. A terminal point

The terminal side of θ passes through $(-7, 24)$.

(a) Find the distance r from the origin and identify the quadrant.

(b) Find the exact values of all six trigonometric functions of θ .